Ref:. Introduction to QM : Chalmers W. Sherwin

The Experimental Basis of QM (Quantum Mechanics)

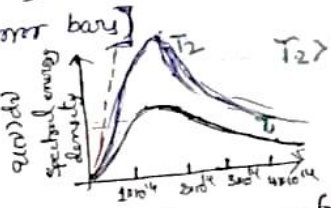
- Before QM, a revolutionary theory is formulated, there is a profusion of experiments; the existing theories fail to explain and deduce scientific phenomena that arose due to challenging experimental results.
- We can identify a small no. of crucial experiments which firmly established the new interpretation of nature.

⇒ Natural phenomena / careful & systematic repetitive experiments
→ propose theoretical models to understand experiments
→ use the theory to further modify iteratively until the theoretical predictions and deductions satisfy the experiments
→ this establishes a theory [There is no "the" theory but there can be the experiment (within experimental error bars)]

Maxwell's theory
Light as Waves was established

NO. 1 experiment for establishing QM

① Black body Radiation: "ONLY the Quantum theory of Light can explain its origin"



$T_2 > T_1$

Hot piece of metal gives off visible light whose color varies with temp. of metal. "R → G → White"
temperature → increase.

→ k_B = Boltzman const = 1.381×10^{-23} J/K (Joule/Kelvin)
→ $\bar{\epsilon}$ = classical average energy = k_BT

--- Rayleigh-Jeans formula

$$\boxed{\text{Rayleigh-Jeans formula} \rightarrow u(\nu)d\nu = \frac{8\pi kT}{c^3}\nu^2 d\nu}$$

Note it①

This is classical physics → about black body radiation

$u(\nu)d\nu$ = energy density in $d\nu \propto \nu^2$
as $\nu \to \infty$, $u(\nu)d\nu \to \infty$

ν → large means, λ small
λ small : $\lambda_{visible} > \lambda_{UV}$

→ Ultraviolet Catastrophe of classical Physics

"QM" (aspect) comes to rescue → Max Planck's "lucky Guess Work"

$$\boxed{\text{Planck Radiation formula} \rightarrow u(\nu)d\nu = \frac{8\pi h}{c^3}\frac{\nu^3 d\nu}{e^{h\nu/k_BT}-1} \rightarrow h = 6.626\times10^{-34}\ J.s.}$$

Student: what happens for $h\nu \gg k_BT$??

Arrhenius

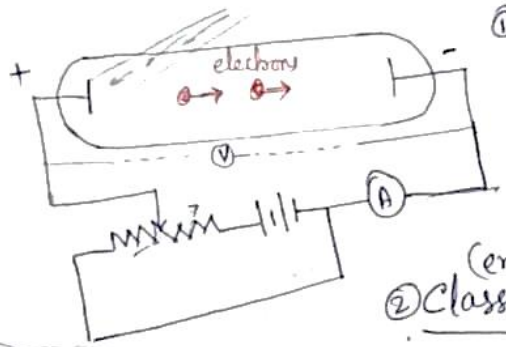
" An oscillator emits radiation of frequency ν when it drops from one energy state to the next lower one, and it jumps to the next higher state when it absorbs radiation of freq ν. Each discrete bundle of $h\nu$ is called a QUANTUM from the Latin word of "how much" "
Average energy: $\epsilon_n = nh\nu$, $\epsilon = h\nu/[\exp(h\nu/kT)-1]$

⇒ Summary of Planck's theory

→ Planck interpreted the experimentally observed curve (Intensity of emitted light vs frequency ; note → Intensity of emitted light) Only by assuming that the electromagnetic radiation was quantized in units of $h\nu$.

"ALL the DISCONTINUITIES in the Nature are meted out in units based directly upon h" The existence of this h and its particular size together form one of the great mysteries of nature. "h" appears explicitly or implicitly in every equation in Quantum theory.

② Photoelectric Effect : "The energies of electron liberated by light depend on the frequency of light."



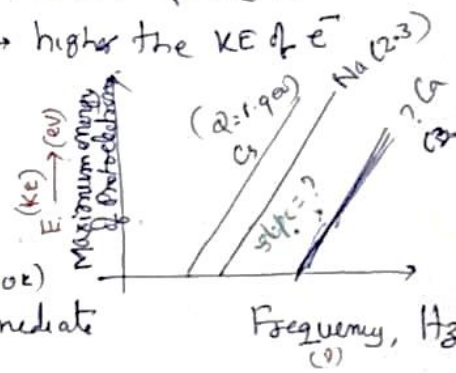
① Classically → Not explainable.
"Within the limits of experimental accuracy (10^{-9} s), there is no time interval between the arrival of light at a metal surface and emission of photoelectrons."

classically means
↳ EM theory

② Classically (em theory) → ① More intense the light, the greater the energies of the e^-.

③ CM fails → higher ν → higher the KE of e^-

Assumption of Einstein → Light is not spread out over wavefront but is concentrated in small packets or photons.



① Spontaneous emission → localization explains (Photon ok)
↳ if light is a wave → spread out → No immediate emission. (No delay in emission)

② Changing "INTENSITY" changes "Photoelectrons" → more intense impinging → more ↑ photoelectrons.

③ higher ν, → higher photon energy.

→ work function, ϕ is minimum energy for an electron to escape from a particular metal surface

· $\phi = h\nu_0$. ν_0 is critical frequency below which no photoelectrons are emitted.

	ϕ
Cs	1.9 eV
Na	2.3 eV
Ca	3.2 eV

P1 $\boxed{\text{Photoelectric effect} : h\nu = KE_{max} + \phi}$

$h\nu$ = incident Photon energy

KE_{max} = maximum Photoe^- energy.

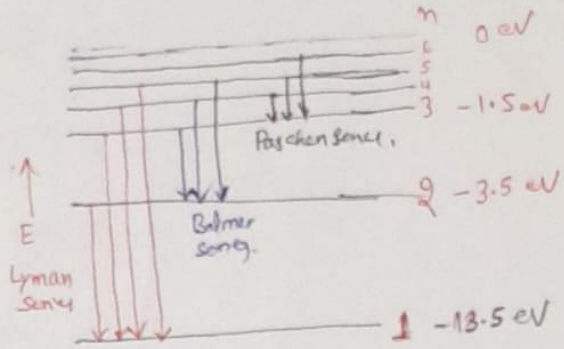
- Black body radiation and Photoelectric effect → Imply the Quantization of Light
  - ↳ that light consist of Photons, a packet of energy $h\nu$ (not a wave concept)

These two showed DISCRETENESS Nature of Light.

③ Atomic Spectra:

Atomic Spectra from gaseous light source showed characteristic discreteness.

Spectral frequency condition
↳ $\Delta E = h\nu$
(Bohr's condition)



Energy levels and the radiative transitions of the hydrogen atom.

④ Quantization of Angular momentum

Bohr's proposal → the angular momentum of the electron about the nucleus was quantized: $\hbar/2\pi$.

Sommerfeld extended ~~by~~ including relativity and provided a quantitative explanation of some of the finer details of the energy levels.

these could not explain: • the orbiting 'Point-electron' should radiate electromagnetically and quickly spiral into nucleus rather than 'jump' downward from one discrete energy level to another, finally settling into a perfectly stable lowest state.

- No explanation of why transitions occurred between certain levels and not between others.
- This theory failed to explain spectra from more complex systems such as those of He and Li.

These are examples → How a theory, although only partially true, will yet produce many qualitative correct predictions.

When $\bar{e}$ are considered as waves → many things could be explained.

Other developments such as

Youngs double slit experiments → showed wave nature of light

X-rays
- Photoelectric effect
- Compton effect } Particle nature
- Diffraction (XRD) : Wave nature

Electrons
- Diffraction
- X-ray emission → EDS (energy dispersive spectroscopic)
- Nowadays → atomic imaging with $\bar{e}$ beam using aberration corrected $\bar{e}$ (typically ≈ 200 - 300 keV $\bar{e}$)

→ Scanning Electron Microscope (SEM)

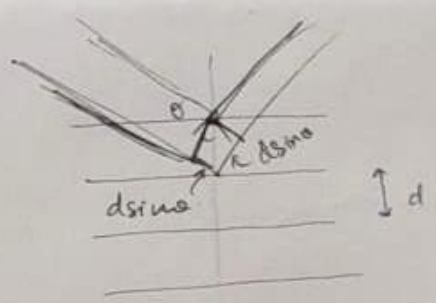
Later developments
- → imaging with secondary $\bar{e}$
- → imaging with secondary x-rays
- → " " cathodoluminescence
- → Backscattered $\bar{e}$ diffraction

→ Transmission electron microscopy

(Later development)
- → Imaging with forward scattered electrons → using waves
- → Imaging with secondary x-rays
- → Imaging with Energy loss of $\bar{e}$
- →

$$\boxed{\text{Bragg's Law} \quad 2d\sin\theta = n\lambda}$$

↳ true for photons
electrons
neutrons
any particle beam



QM Basics

- In QM we deal with Waves.
- Waves are described by wave functions, denoted by ψ. → complex < real / imaginary
- ψ → itself has no physical interpretation → $\psi \to f(x,t) \to 1d$
  $\to f(r,t) \to 3d$
- $|\psi|^2$ → evaluated at a specific $\bar{r}$ and t,
  is proportional to the probability of finding the body there at that time.
- p, L (angular momentum), E can be established from ψ.

⇒ QM → determines ψ for a body when its freedom of motion is limited by the action of external forces.

- ψ is complex, but $|\psi|^2$ is +ve real quantity
- ψ^* is complex conjugate.

$\psi = A + iB$, A & B are real

$\psi^* = A - iB$ complex conjugate

$\therefore |\psi|^2 = \psi^*\psi = (A+iB)(A-iB) = A^2 - i^2B^2 = A^2 + B^2$ $[i^2 = -1]$

What are the qualities of a good ψ? $\int_{-\infty}^{+\infty} |\psi|^2 dV \neq 0 \Rightarrow \int_{-\infty}^{+\infty} |\psi|^2 dV = 1$.

Particle exists somewhere at all time.

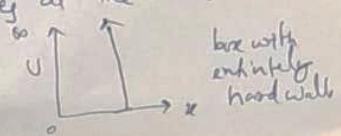
ψ is said to be normalized → $\int_{-\infty}^{+\infty} |\psi|^2 dV = 1$. ⇒ $P_{x_1 x_2} = \int_{x_1}^{x_2} |\psi|^2 dx$

$\int_{-\infty}^{+\infty} P\, dV = 1$

Well behaved wave functions

1. ψ must be continuous and single-value everywhere.
2. $\frac{\partial\psi}{\partial x}, \frac{\partial\psi}{\partial y}, \frac{\partial\psi}{\partial z}$ must be continuous and single-value everywhere.
3. ψ must be normalizable, $\int_{-\infty}^{+\infty} |\psi|^2 dV$ has a ~~finite~~ finite const.

These rules are not always obeyed by the ψ's in model situations that only approximate actual ones: ψ for a particle in a box with infinitely hard walls do not have continuous derivatives at the walls, since $\psi = 0$ outside the box;



Wave Equation

Let y propagate in the x direction with a speed v.

Stretched Spring → y → displacement of the string from x-axis.

Sound → y is pressure difference.

Light → y is either electric or magnetic field magnitude.

Particle → Meet here ← Waves

Particle

1. $p = mv$
2. $KE = \frac{1}{2} m v^2 = \frac{(mv)^2}{2m} = \frac{p^2}{2m}$
3. Total $E = KE + PE \rightarrow E = \frac{p^2}{2m} + V$
4. $F = ma = m\frac{d^2x}{dt^2}$

$\lambda = \frac{h}{p}$

$k = \frac{2\pi}{\lambda}$

$p = \frac{h}{\lambda} = \frac{hk}{2\pi}$

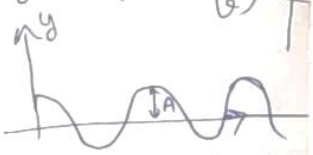
$p = \hbar k$

Waves

$F = ma$

$$\boxed{\frac{d^2y}{dx^2} = \frac{1}{v^2}\frac{d^2y}{dt^2}}$$

$y = F\left(t \pm \frac{x}{v}\right)$



$y = A\cos\omega\left(t - \frac{x}{v}\right)$

$y = F\left(t \pm \frac{x}{v}\right)$

Free Particle: a particle that is not under influence of any force and pursues a straight path at constant speed.

$$y = Ae^{-i\omega(t - x/v)} \Rightarrow \psi = Ae^{-i\omega(t - x/v)}, \quad \psi^* = Ae^{+i\omega(t - x/v)}$$

$$e^{-i\theta} = \cos\theta - i\sin\theta \Rightarrow y = Ae^{-i\omega(t - x/v)}$$

$$y = A\cos\omega(t - x/v) - iA\sin\omega(t - x/v).$$ → for a stretched string → imaginary → 0

· Does $y = Ae^{-i\omega(t - x/v)}$ satisfy wave equation? $y = e^{ix}$

$$\frac{d}{dx}(e^{ix}) = e^{ix}\frac{dy}{dx} \Rightarrow y = e^{-i\omega(t - x/v)} \Rightarrow \frac{\partial y}{\partial x} = -i\omega(-1/v)\,y = \frac{+i\omega}{v}y$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial y}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{-i\omega}{v}y\right) = \left(\frac{-i\omega}{v}\right)^2 y = \frac{i^2\omega^2}{v^2}y = -\frac{\omega^2}{v^2}y$$

$$\partial y/\partial t = -i\omega y, \quad \partial^2 y/\partial t^2 = i^2\omega^2 y = -\omega^2 y \quad \Big| \quad \frac{\partial^2 y}{\partial x^2} = (-\omega^2)\frac{1}{v^2}y = -\frac{\omega^2 y}{v^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

Schrodinger equation :

$$F = m\frac{d^2x}{dt^2} \Rightarrow \frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

For elastic waves

$$\rho\frac{\partial^2 y}{\partial t^2} = Y\frac{\partial^2 y}{\partial x^2}$$

$$\frac{d^2y}{dx^2} = \left(\frac{\rho}{Y}\right)\frac{d^2y}{dt^2} = \frac{1}{v^2}\left(\frac{d^2y}{dt^2}\right)$$

$$\frac{1}{v^2} = \frac{\rho}{Y} \Rightarrow v^2 = Y/\rho \text{ or } v = \sqrt{Y/\rho}$$

$$\psi = A e^{-i\omega(t - x/v)}$$

$\omega = 2\pi\nu$ and $v = \lambda\nu$

$$\psi = A e^{-i2\pi\nu(t - x/\lambda\nu)} = A e^{-2\pi i(\nu t - x/\lambda)}$$

$$\boxed{\therefore \psi = A e^{-2\pi i(\nu t - x/\lambda)}}$$

→ our Job now is to get p, E, x.

$E = h\nu$, $\lambda = \frac{h}{p} = \frac{2\pi\hbar}{p}$ $\quad (\hbar = h/2\pi)$

$\nu = \frac{E}{h}$; $\frac{1}{\lambda} = p/2\pi\hbar$

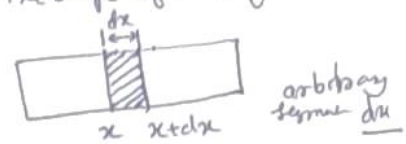
$$\psi = A e^{-2\pi i\left(\frac{E}{h}t - \frac{x \cdot p}{2\pi\hbar}\right)}$$

$$= A e^{-\frac{2\pi}{h}i(Et - px)}$$

$$E = A e^{-(i/\hbar)(Et - px)}$$

Elastic Waves

Propagation of an elastic wave in the shape of a long bar:



Longitudinal wave propagation.

elastic displacement, (at the point) $y(x)$.

~~Stress strain~~ Strain $= \frac{dy}{dx}$

$\varepsilon = \frac{dy}{dx}$ is change per unit length.

S, Stress is Force/Area

$$\frac{S}{\varepsilon} = Y \text{ or } S = Y\varepsilon$$

$$S = Y\frac{dy}{dx} \Rightarrow S = \frac{F}{A}$$

$$F = ma = (\rho A' dx)\frac{\partial^2 y}{\partial t^2}$$

is Change in Stress

$$\underbrace{[S(x+dx) - S(x)]}_{(\partial S/\partial x)dx} A'$$ → for short segment

$$(\rho A' dx)\left(\frac{\partial^2 y}{\partial t^2}\right) = \left(\frac{\partial S}{\partial x}\right)A' dx$$

↳ $S = Y\frac{dy}{dx}$

$$\frac{\partial S}{\partial x} = \frac{\partial}{\partial x}\left(Y\frac{dy}{dx}\right)$$

$$\frac{\partial S}{\partial x} = Y\frac{d^2y}{dx^2}$$

$\therefore F = ma \Rightarrow$

$$\rho(A')dx\frac{\partial^2 y}{\partial t^2} = Y\frac{d^2y}{dx^2}$$

$$\frac{d^2y}{dx^2} = \frac{\rho}{Y}\frac{d^2y}{dt^2} \Rightarrow \left(\frac{1}{v^2} = \frac{\rho}{Y}\right)$$

Free particle, a wave equivalent of an unrestricted particle of total energy E and momentum p moving in x + ve direction.

↳ a harmonic displacement wave moving freely along a stretched string. $\boxed{\psi = A e^{-(i/\hbar)(Et - px)}}$

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2} \Rightarrow \frac{\partial^2 \psi}{\partial x^2} = ?$$

$$\frac{\partial \psi}{\partial x} = (-i/\hbar)(-p) = +\frac{i}{\hbar}p$$

$$\frac{\partial^2 \psi}{\partial x^2} = \left(+\frac{i}{\hbar}p\right)\left(+\frac{i}{\hbar}p\right)\psi = i^2\frac{p^2}{\hbar^2}\psi = -\frac{p^2}{\hbar^2}\psi$$

$$\therefore \boxed{p^2\psi = -\hbar^2\frac{\partial^2\psi}{\partial x^2}}$$

$$\frac{\partial \psi}{\partial t} = (-i/\hbar)(E)\psi \Rightarrow \boxed{E\psi = -\frac{\hbar}{i}\frac{\partial \psi}{\partial t}}$$

$$E = \frac{p^2}{2m} + U(x) \Rightarrow E\psi = \frac{p^2\psi}{2m} + U(x)\psi$$

$$\therefore -\frac{\hbar}{i}\frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + U(x)\psi$$

$$\therefore \frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + i\hbar\frac{\partial\psi}{\partial t} - U(x)\psi = 0.$$

(3d) $$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2}\right) + U\psi.$$

Linearity and Superposition

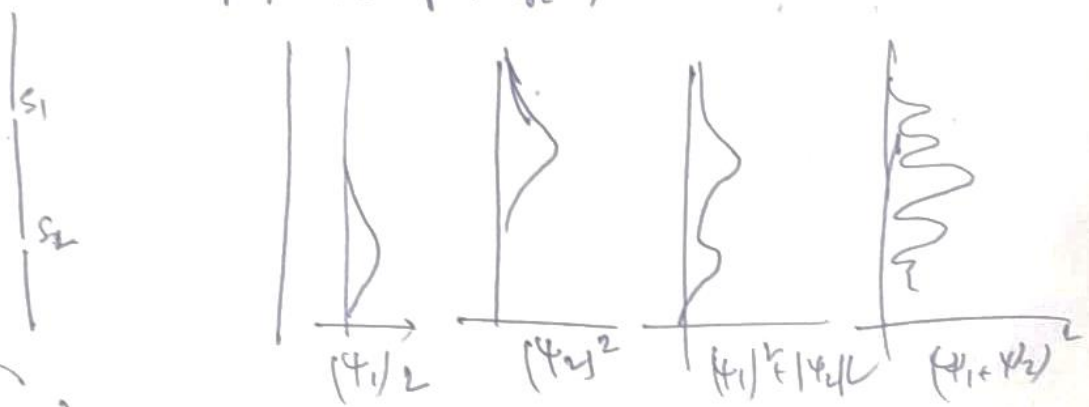
SE → linear in ψ.
↳ ψ and its derivatives

No terms independent of ψ or that involve higher powers of ψ or its derivatives.

ψ_1 and ψ_2 are solutions.

$\psi = a_1\psi_1 + a_2\psi_2$ is also a solution.

Superposition principle.



$(\psi = \psi_1 + \psi_2)$

$P_1 = |\psi_1|^2, \quad P_2 = |\psi_2|^2$

$$\therefore P = |\psi|^2 = |\psi_1 + \psi_2|^2 = (\psi_1^* + \psi_2^*)(\psi_1 + \psi_2)$$

$$= \psi_1^*\psi_1 + \psi_2^*\psi_2 + \psi_1^*\psi_2 + \psi_2^*\psi_1$$

$$= P_1 + P_2 + \psi_1^*\psi_2 + \psi_2^*\psi_1$$

Operator

$$\frac{\partial \psi}{\partial x} = \frac{i}{\hbar} p\psi \Rightarrow p\psi = \frac{\hbar}{i}\frac{\partial}{\partial x}\psi$$

$$\frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} E\psi \Rightarrow E\psi = i\hbar \frac{\partial}{\partial t}\psi,$$

$$\hat{p} = \frac{\hbar}{i}\frac{\partial}{x}, \quad \hat{E} = i\hbar\frac{\partial}{\partial t}$$

$$\hat{E} = \hat{KE} + \hat{U} = \frac{p^2}{2m} + U$$

$$KE = \frac{p^2}{2m} = \frac{1}{2m}\left(\frac{\hbar}{i}\frac{\partial}{\partial x}\right)^2 = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$$

$$\therefore i\hbar\frac{\partial}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + U$$

$$\text{OR} \quad i\hbar\frac{\partial}{\partial t}\psi = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + U(\psi)$$

$$\langle p\rangle = \int \psi^* \hat{p}\psi = \int \psi^* \left(\frac{\hbar}{i}\frac{\partial}{x}\right)\psi = \frac{\hbar}{i}\int \psi\frac{\partial\psi}{\partial x}dx$$

$$\langle E\rangle = \int \psi^* E\psi = \int \psi^* \left(i\hbar\frac{\partial}{\partial t}\right)\psi = i\hbar\int \psi^*\frac{\partial\psi}{\partial t}dx$$

$$\langle G(x,p)\rangle = \int_{-\infty}^{\infty} \psi^* \hat{G}\psi\, dx.$$

· SE → Steady form

time Pos.

$$\Psi = A e^{-(i/\hbar)(Et - px)} = A e^{-(iE/\hbar)t} \cdot e^{+(ip/\hbar)x}$$

$$\Psi = \underbrace{\psi e^{-(iE/\hbar)t}}_{1}, \quad \text{where } \psi = e^{(+ip/\hbar)x}$$

$$E\Psi = \frac{p^2}{2m}\Psi + U\Psi$$

$$\frac{E.\psi e^{-(iE/\hbar)t}}{e^{-(iE/\hbar)t}} = \frac{-\frac{\hbar^2}{2m} e^{-(iE/\hbar)t} \frac{\partial^2\psi}{\partial x^2} + U\psi e^{-(\frac{iE}{\hbar})t}}{e^{-(iE/\hbar)t}}$$

$$E\psi = \frac{-\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + U\psi$$

$$\therefore \quad -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} = (E - U)\psi$$

$$\text{or } \frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + (E - U)\psi = 0$$

$$\frac{\partial^2\psi}{\partial x^2} + \frac{2m}{\hbar^2}(E - U)\psi = 0$$

Each wave has a specific ω

Introduction to QM by Griffiths

Lecture · B Tech (II) 8/6/23

①

QM → ~~Postulations~~ The postulates of QM for a system consisting of two particles $(x_1, y_1, z_1 ; x_2, y_2, z_2 ; t)$

① The wave function, ψ, belonging to a system composed of two particles, is a function of $x_1, y_1, z_1, x_2, y_2, z_2$, and t.

② The wave function for ψ is obtained from the classical expression for the total energy of system by replacing the dynamical variables by

Dynamical Variable	Operator
x_1 (position)	x_1
P_{x_1} (momentum)	$\frac{\hbar}{i}\frac{\partial}{\partial x}$
E (total Energy)	$-\frac{\hbar}{i}\frac{\partial}{\partial t}$ $\left(i\hbar\frac{\partial}{\partial t}\right)$

} similarly for y_1, z_1, x_2, y_2, z_2

and by the insertion of the wavefunction ψ as the operand,

③ ψ, and $\frac{\partial\psi}{\partial x_1}, \frac{\partial\psi}{\partial y_1}, \ldots \frac{\partial\psi}{\partial z_2}$ must be everywhere continuous, finite, and single valued, throughout the configuration space (the full range of all six variables).

④ ψ is normalized, i.e. $\int_{\text{all configuration space}} \psi^* \psi \, dx_1 dy_1 dz_1 dx_2 dy_2 dz_2 = 1$

The expectation value of the observable, α is calculated from the wave function by the formula...

$$\hat{\alpha} = \int_{\text{all conf space}} \psi^* \alpha_{(\text{operator})} \psi \, dx_1 dy_1 dz_1 dx_2 dy_2 dz_2$$

Wave function

$\psi = A + iB$ A & B are real functions

$\psi^* = A - iB$

$\therefore |\psi|^2 = \psi^*\psi = A^2 - i^2B^2 = A^2 + B^2$

$\int |\psi|^2 dV = 1 \implies \int P\, dV = 0$

For a particle restricted to motion in x-direction

$P_{x_1, x_2} = \int_{x_1}^{x_2} |\psi|^2 dx$.

expectation Values

$\langle x \rangle = \int_{-\infty}^{\infty} x |\psi|^2 dx$

Prob

Exe: $\psi = ax$ for $1 \le x \ge 0$.

what is the probability the particle can be found between $x = 0.4$ and $x = 0.6$.

Find the expectation value of $\langle x \rangle$ of the particle Position

Ans: Prob: $\int_{x_1}^{x_2} |\psi|^2 dx = \int_{x_1}^{x_2} (ax)^2 dx = a^2 \left[\frac{x^3}{3}\right]_{0.4}^{0.6}$

for 0.45 to 0.55 → $\frac{1}{3}\left[(0.6)^3 - (0.4)^3\right] = \frac{a^2}{3}[0.216 - 0.064]$

$= \frac{0.152}{3} a^2$

$a = 3 \rightarrow \psi = 3x; \rightarrow$ Prob: $\int_{0.4}^{0.6} 3x = 0.0506 \times 3 = 0.152$

expation: $\langle x \rangle = \int_0^1 x |\psi|^2 dx = \int_0^1 x \cdot 9x^2 dx = 9\int_0^1 x^3 dx = 27 \left[\frac{x^4}{4}\right]_0^1 = \frac{9}{4} = 2.25$

Wave equation

$\boxed{\psi(x;t)}$

$$E\psi = \frac{p^2}{2m}\psi + V\psi$$

$$-\frac{\hbar}{i}\frac{\partial}{\partial t}\psi = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)$$

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi$$

Separation Variably

$$\psi(x,t) = \psi(x)\,\phi(t)$$

$$\frac{\partial\psi}{\partial t} = \psi\frac{d\phi}{dt}, \quad \frac{\partial^2\psi}{dx^2} = \frac{d^2\psi}{dx^2}\phi$$

$$i\hbar\,\psi\frac{d\phi}{dt} = -\frac{\hbar^2}{2m}\cdot\frac{d^2\psi}{dx^2}\phi + V\psi\phi$$

devide by $\psi\phi$

$$\frac{i\hbar\,\cancel{\psi}\frac{d\phi}{dt}}{\cancel{\psi}\phi} = \frac{-\hbar^2}{2m\psi\phi}\frac{d^2\psi}{dx^2}\cdot\phi + \frac{V\psi\phi}{\psi\phi}$$

$$i\hbar\frac{1}{\phi}\frac{d\phi}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\psi}\frac{d^2\psi}{dx^2} + V$$

this 't' dep.

this is x fn.

Let $i\hbar\frac{1}{\phi}\frac{d\phi}{dt} = E$

$$\frac{d\phi}{dt} = \frac{1}{i\hbar}\phi E$$

$$\frac{d\phi}{dt} = -\frac{iE}{\hbar}\phi$$

Separation const.

$$i\hbar \frac{1}{\phi}\frac{d\phi}{dt} = E$$

separation const

Operator $\longrightarrow$ $E = -\frac{\hbar}{i}\frac{\partial}{\partial t}$

$\psi\phi = \psi\phi$

$$E\psi = -\frac{\hbar}{i}\frac{\partial \psi}{\partial t} = -\frac{\hbar}{i}\frac{\partial(\psi\phi)}{\partial t}$$

$$E\psi\phi = -\frac{\hbar\psi}{i}\frac{\partial\phi}{\partial t}$$

$$E = \frac{-\hbar}{i\phi}\frac{d\phi}{dt}$$

$$\frac{d\phi}{dt} = -\frac{iE}{\hbar}\phi$$

$$\boxed{-\frac{\hbar^2}{2m}\frac{1}{\psi}\frac{d^2\psi}{dx^2} + V = E}$$

$$E = -\frac{\hbar^2}{2m}\frac{1}{\psi}\frac{d^2\psi}{dx^2} + V$$

$$E\psi = -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V$$

$\Rightarrow$ Two differential eq

$$\frac{d\phi}{dt} = -\frac{iE}{\hbar}\phi \quad (1)$$

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi = E\psi} \quad (2)$$

$$\frac{d\phi}{dt} = -\frac{iE}{\hbar}\phi \Rightarrow \phi(t) = e^{-iEt/\hbar}$$

$$\boxed{\psi(x,t) = \psi(x)\, e^{-iEt/\hbar}}$$

<u>Probability</u>

$$|\psi(x,t)|^2 = \psi^*\psi$$

$$|\psi|^2 = \psi^* e^{+iEt/\hbar}\, \psi\, e^{-iEt/\hbar} = |\psi|^2$$

Every expection value is comp in time.

<u>Hamiltonian</u> = total energy

$$H(x,p) = \frac{p^2}{2m} + V(x)$$

Hamiltonian Operator, ~~Subtrans~~.

$$p \to -i\hbar\frac{\partial}{\partial x}$$

$$H(x,p) \quad \boxed{\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)}$$

$$\therefore \quad \hat{H}\psi = E\psi$$

$$\langle H\rangle = \int \psi^* \hat{H}\psi\, dx = E\int |\psi|^2 dx = E$$

$$\langle H^2\rangle = \hat{H}(\hat{H}\psi) = \hat{H}(E\psi) = E(\hat{H}\psi) = E^2\psi$$

$$\therefore \quad \langle H^2\rangle = \int \psi^* \hat{H}^2 \psi\, dx = E^2 \int |\psi|^2 dx = E^2$$

Variance of H ?

(6)

$$\sigma_H^2 = \langle H^2 \rangle - \langle H \rangle^2 = E^2 - E^2 = 0 \,!$$

<u>Conclusion:</u> A separable solution has the property that every measurement of total energy is certain to return to 'E'.

<u>Linear combination</u>

we get $\psi_1(x), \psi_2(x), \psi_3(x), \ldots$ we get infinite collection.

$\{\psi_n(x)\}$ — each with allowed energies (E_1, E_2, E_3)

$$\Psi_1(x,t) = \psi_1(x)\, e^{-iE_1 t/\hbar}$$

$$\Psi_2(x,t) = \psi_2(x)\, e^{-iE_2 t/\hbar}$$

$$\therefore \Psi(x,t) = \sum_{n=1}^{\infty} c_n \psi_n(x)\, e^{-iE_n t/\hbar}$$

Summary :

We have $V(x)$ $\underline{\underline{\text{not}}}$ $V(x, t)$

→ Starting wavefunction : $\psi(x, 0)$

→ Your job is to find the wavefunction $\psi(x, t)$.
for any subsequent time, t.

→ this means we need to solve <u>time dependent</u> SE.

<u>How</u>) (i) Solve time indep. → ~~solution~~
↳ this gives an infinite set of solutions,
$\{\psi_n(x)\}$ — each with $\{E_n\}$.

$$\psi(x, 0) = \sum_{n=1}^{\infty} c_n \psi_n(x)$$

Now $\psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar} = \sum_{n=1}^{\infty} c_n \psi_n(x, t)$.

$$\therefore \boxed{\psi_n(x, t) = \psi_n(x) e^{-iE_n t/\hbar}}$$

Problem

Problem: Suppose a particle starts out in a linear combination of just ~~two particles~~ two stationary states: $\Psi(x,0) = c_1\psi_1(x) + c_2\psi_2(x)$ (9)

What $\Psi(x,t)$ at subsequent time?

Find the probability density and describe its motion?

$$\Psi(x,t) = c_1\psi_1(x)e^{-iE_1t/\hbar} + c_2\psi_2(x)e^{-iE_2t/\hbar}$$

where E_1 and E_2 are the energies associated with ψ_1 & ψ_2.

Probability density: $|\Psi(x,t)|^2 = \left(c_1\psi_1 e^{+iE_1t/\hbar} + c_2\psi_2 e^{+iE_2t/\hbar}\right)\left(c_1\psi_1 e^{-iE_1t/\hbar} + c_2\psi_2 e^{-iE_2t/\hbar}\right)$

$$= c_1^2\psi_1^2 + c_2\psi_2^2 + 2c_1c_2\psi_1\psi_2\cos\left[(E_2-E_1)t/\hbar\right].$$

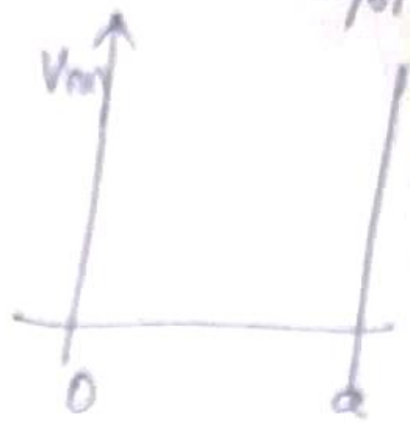
→ Probability density oscillates sinusoidally at an angular frequency $\omega = (E_2-E_1)/\hbar$.

This is NOT a stationary state

But linear combination of stationary states to produce motion

① Infinite Square well 8/6/25

$$V(x) = \begin{cases} 0 & 0 \le x \le a \\ \infty & \text{otherwise} \end{cases}$$



$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$$

or $$\frac{d^2\psi}{dx^2} = -k^2\psi \quad , \quad k = \sqrt{\frac{2mE}{\hbar}}$$

$$\psi(x) = A\sin kx + B\cos kx$$

For above potential, $\boxed{\psi(x) = A\sin kx}$

$$ka = 0, \pm\pi, \pm 2\pi$$

$$ka = \frac{n\pi}{a} \quad , \text{ for with } n = 1, 2$$

$$\frac{n\pi}{a} = \sqrt{2mE/\hbar^2}$$

$$\frac{n^2\pi^2}{a^2} = \frac{2mE}{\hbar^2} \Rightarrow E = \frac{n^2\pi^2\hbar^2}{2ma^2}$$

$$E_n = \frac{n^2h^2}{8ma^2}$$

$$\int_0^a |A|^2\sin^2(kx)\,dx = |A|^2\frac{a}{2} = 1 \quad , \quad |A|^2 = 2/a$$

$$A = \sqrt{2/a}$$

$$\psi_n(x) = \sqrt{2/a}\,\sin\left(\frac{n\pi}{a}x\right)$$

What happens :

$$\int \psi_m^*(x)\, \psi_n(x)\, dx = ??$$

$\longrightarrow 0$. Check

$$\int \psi_m^*(x)\, \psi_n(x)\, dx = \delta_{mn}.$$

$$\delta_{mn} = \begin{cases} 0 & m \neq n \\ 1 & m = 1 \end{cases}$$

$\hookrightarrow$ Kronecker delta.

ψ_i are orthonormal.

$$\Psi_n(x,t) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-\frac{iEt}{\hbar}}$$

$$\Psi_n(x,t) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a} x\right) e^{-i(n^2\pi^2\hbar/2ma^2)t}$$

general solution is a linear combination

$$\Psi(x,t) = \sum_{n=1}^{\infty} c_n \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a} x\right) e^{-i(n^2\pi^2\hbar/2ma^2)t}$$

$$\Psi(x,0) = \sum_{n=1}^{\infty} c_n \psi_n(x)$$

$$c_n = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{n\pi x}{a}\right) \Psi(x,0)\, dx.$$

Example : A particle in the infinite square well ~~A Part~~ has the initial wave function

$$\psi(x,0) = \underline{A x (a-x)}, \quad (0 \le x \le a)$$
$$= 0 \quad \text{elsewhere},$$

Find $\psi(x,t)$.

What is A ? $\int_0^a |\psi(x,0)|^2 dx = 1$

$$|A|^2 \int_0^a x^2 (a-x)^2 dx = |A^2| \frac{a^5}{30}$$

$$\therefore \quad A = \sqrt{\frac{30}{a^5}}$$

$$c_n = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{n\pi}{a} x\right) \sqrt{\frac{30}{a^5}}\, x (a-x)\, dx.$$

$$= \begin{cases} 0 & n \text{ even} \\ 8\sqrt{15}/(n\pi)^3 & n \text{ odd.} \end{cases}$$

$$\therefore \psi(x,t) = \sqrt{\frac{30}{a}} \left(\frac{2}{\pi}\right)^3 \sum \frac{1}{n^3} \sin\left(\frac{n\pi}{a} x\right) e^{-i n^2 \pi^2 \hbar t / 2ma}$$

Harmonic oscillator

$$F = -kx = -m\frac{d^2x}{dt^2}$$

$$x(t) = A\sin(\omega t) + B\cos(\omega t)$$

$$\omega = \sqrt{k/m}$$

$$V(x) = \tfrac{1}{2}kx^2$$

$$V(x) = V(x_0) + V'(x_0)(x-x_0) + \frac{1}{2}V''(x_0)(x-x_0)^2 + \dots$$

$$V = \frac{1}{2}m^2\omega^2x^2$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2x^2\psi = E\psi$$

$$E_n = \left(n+\frac{1}{2}\right)\hbar\omega$$

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}$$

$$E_0 = \tfrac{1}{2}\hbar\omega$$